BrickPlan: A Benchmark for Visual Assembly and Error Recovery

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Figure 1. **Repair a supplied state or keep building through an unexpected change.** The collection has 5,720 distinct task records. Its 5,452 main candidates comprise 3,697 static tasks and 1,755 dynamic tasks; 268 historical dynamic structure/crop tasks remain diagnostic only. Dynamic object tasks cover 50 whole-object and 13 named-module configurations. Small original LDraw objects retain every source part; larger objects supply the surrounding assembly.

Abstract

Visual assembly requires an agent to maintain a correct structure while its own actions and unexpected changes alter the scene. We introduce BRICKPLAN*, a benchmark for static repair and continuous construction with hidden interventions. The organized collection contains 5,720 distinct tasks, including 5,452 main candidates spanning local static structures, complete objects, and named remaining modules. Dynamic tasks cover 63 operation configurations and include 43 episodes with two or three interventions. Seven physical error labels are distinguished from access-related process errors and intervention counts. Agents inspect actual RGB states and execute one brick action at a time; acceptance compares terminal geometry, color, and part type, allowing different legal construction orders. We define static correction and damage metrics, and dynamic recovery metrics with explicit control, exposure, and event-effect requirements. All 5,922 retained execution conditions pass reference replay, while a historical route diagnostic shows that reference-prefix triggers miss opportunities on valid alternative routes. These are executor checks, not learned-agent results. We provide named model candidates and unmeasured experiment tables; current-interface human qualification and formal model evaluation remain outstanding.*

1. Introduction

An agent assembling a toy car may build the left side before the right, while another follows the opposite order. Both can finish correctly. An evaluator that compares every intermediate state with one manual would reject a valid route. Scheduling an accident only when an agent reproduces that same reference prefix creates a second problem: a correct alternative route may never encounter the accident. A benchmark must separate the final structure, physical action legality, and actual exposure to a fault.

The visual specification also matters. A final image can hide joints or internal pieces, making reconstruction ambiguous even for a person. We therefore distinguish local static repair from dynamic construction of recognizable objects. Small community models are built in full; large scenes supply an already assembled body and ask the agent to complete an identifiable roof, cabin, or tail. Available manuals describe one feasible route. They establish in-

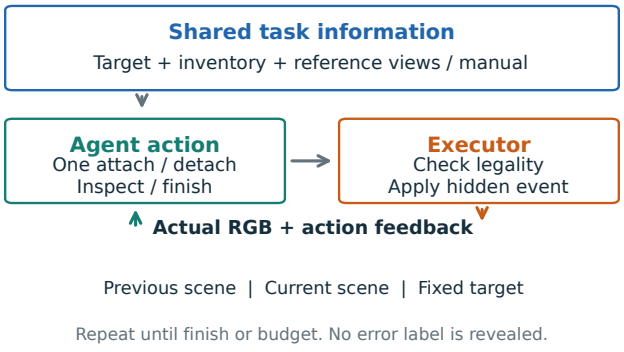

Figure 2. **Continuous interaction.** One target remains available. Each action is checked; the next observation shows the actual state after execution and any hidden intervention. An error label or correction is never supplied. Manuals are queried where supported.

tended structure without imposing that route as the only acceptable sequence.

BRICKPLAN connects these settings through a common interaction contract. A static task supplies an imperfect state and one stage target. A dynamic task supplies a fixed whole-object or module target, then allows uninterrupted inspection and one-brick operations. Hidden interventions can remove, replace, move, or add pieces after progress has been made. The agent must finish the task through ordinary actions, without a forced mid-episode declaration that an error occurred.

Prior work addresses buildable generation [1, 5], manual execution [10], assembly-step correction [9], and interactive structural understanding [8]. Agentic design and robot recovery further motivate robust evaluation [2, 3]. Our contribution is the task and measurement contract linking visual history, legal alternative orders, and externally altered ongoing work.

Contributions. (1) We organize 5,720 distinct tasks with explicit scope, source grouping, error labels, and structural grades; 5,452 are main candidates. (2) We define a shared visual action interface and terminal scoring that accepts interchangeable parts and alternative legal orders. (3) We separate correction from damage, and raw completion from qualified recovery, including repeated interventions and technical censoring. Reference replays and route diagnostics support these protocol claims. Formal model rankings and human performance are not yet established.

2. Related Work

Brick generation and visual assembly. BrickGPT generates buildable structures using validity filtering and rollback [5]; BrickNet models connector graphs and assembly sequences from human-designed LDraw objects [1]. BrickBench evaluates agentic design under validity, alignment, and quality criteria [2]. We reuse brick geometry and connection infrastructure rather than claim a new physics simulator. Our goal is specified in advance, so correctness is restoration or completion of that goal rather than open-ended design quality.

Break and Make combines inspection, disassembly, and reconstruction [8]. LEGO-Puzzles and PhyBlock study spatial reasoning, planning, and physical understanding through block tasks [4, 6]. MEPNet executes visual manuals [10]; LEGO-ECA and SCANet address errors and corrections associated with assembly steps [9]. These are close to our static condition. Our dynamic condition retains the agent's ongoing history across an external change and permits inspection, repair, and further construction in the same episode.

Adaptation and recovery. AsgardBench evaluates visually grounded plan adaptation with lightweight feedback [7]; RoboRecover evaluates continuation from execution deviations [3]. Recovery itself is therefore not novel. Our distinction is instruction-assisted brick construction with one-part execution, alternative-order acceptance, and recovery eligibility conditioned on what actually happened. We adapt part-quality and correction/damage analysis [8, 9], while defining metrics for our own action and observation contract.

3. Dataset Construction and Organization

3.1. Counting tasks, conditions, and objects

Table 1 separates task scopes. The catalogue contains 5,839 records: 5,793 earlier records plus 46 conditions for three complete LDraw objects. Within the same source, configuration, and task contract, physical equivalence identifies 119 repeated records: 27 static problems and 92 normal dynamic controls. Deduplication leaves 5,720 distinct tasks. This is not a claim of global semantic deduplication across different sources. An additional 83 cross-adapter aliases remain execution conditions, not new catalogue tasks. Thus 5,922 conditions are retained for replay.

The main candidate set contains 3,697 static and 1,755 dynamic tasks. Static local fragments remain useful for structure repair. In contrast, 224 historical dynamic mechanism tasks and 44 arbitrary-crop dynamic tasks are excluded from object-construction conclusions. They remain available as separate diagnostics. "Candidate" denotes program-checked content; it does not certify that every image has passed independent human qualification.

Table 1. **Current dataset after within-contract deduplication.** Controls are included. Main candidates sum to 5,452; the two diagnostic rows add 268. Configuration counts and task counts are different units, and source objects overlap across static and dynamic tasks.

Task scope	Distinct tasks	Main	Start and acceptance target
Local static structure	3,315	Yes	Supplied local state → its required stage.
Whole-object static stage	148	Yes	Supplied whole-object stage → stage target.
Named-module static stage	234	Yes	Supplied body and module state → stage target.
Whole-object dynamic	1,534	Yes	Empty white plate → complete object (50 configurations).
Named-module dynamic	221	Yes	Supplied body → completed named module (13 configurations).
Historical dynamic structure	224	No	Structural execution and recovery diagnostics.
Archived arbitrary-crop dynamic	44	No	Traceability only; excluded from object main results.
Total	5,720	5,452	3,697 main static tasks and 1,755 main dynamic tasks.

3.2. Recognizable objects and retained coverage

The 63 primary dynamic configurations comprise 50 whole-object builds and 13 named-module builds. Whole objects include all 42 earlier procedural designs, five longer procedural objects, and three complete original LDraw models. The latter are a 12-part ship (4018), a 13-part sea helicopter (4017), and a nine-part helicopter (4037). Their downloaded source files retain every part, color, and transform; none is removed to simplify the task. This certifies source-file coverage, not a retail-set bill of materials. The five longer builds contain 21–32 active parts.

Named modules come from twelve community source scenes, with an additional expanded museum-roof configuration. Their surrounding scene is supplied as immutable context. These are thirteen operation configurations, not thirteen new source sets. Earlier chairs, tables, cars, and machines remain in the suite. The supplement illustrates all dynamic configurations and enumerates all 160 configurations in the original static bank.

3.3. Errors, process demands, and structural grades

Seven basic labels describe physical discrepancies: missing, extra, wrong color, wrong part, displaced position/height, non-equivalent orientation, and swapped positional relations. Labels may co-occur. Part substitutions include size and thickness; local extra layers are extra-piece proxies, not whole-model layer duplication.

Process tags describe premature covering and buried repair, rather than adding more mutually exclusive physical classes. A buried color error can carry both the color and buried-repair labels. Multiple simultaneous defects differ from multiple temporally separated accidents. The 43 main repeated-event tasks contain 37 two-event and six three-event conditions. Community-inspired human-error tags cover twelve approved source categories; they describe synthetic analogues, not measured frequencies or collected human mistake trajectories.

We use separate structural descriptors. Build length is B1 ($\leq$ 12), B2 (13–24), or B3 ($\geq$ 25) active target pieces. Static length counts only its acceptance stage. Rework is R0 for controls; R1 for a reference repair of at most two moves without removing correct covers; R2 for remaining ordinary/multiple-defect repair; and R3 when the witness removes correct covers or uses at least seven moves. These are feasible-witness burdens, not proven minimum requirements. T0–T3 count external events; all static tasks are T0. Visibility is recorded independently. Planning grades G1/G2/G3 combine these descriptors and contain 2,577/2,823/52 main candidates. They are not empirically validated "easy/medium/hard" labels. The supplement gives the exact rules.

Table 2. **Physical error coverage in main candidates.** S: static; W/M: whole/module dynamic. Labels co-occur; rows do not sum to task totals. Module swap is an explicit coverage gap.

Physical label	S	W	M
E01 Missing	1,361	209	60
E02 Extra	356	167	36
E03 Color	1,458	426	60
E04 Part	339	185	15
E05 Position	804	227	32
E06 Orientation	489	130	25
E07 Swap	31	20	0

4. Observation and Execution Contract

4.1. One acceptance target and one action object

A task has fixed context B, movable inventory P, finite pose candidates C_p, initial state s_0, and target g. A part is installed at one candidate or remains on the rack. All tasks use a fixed construction coordinate frame and white-baseplate environment; module tasks also retain a preassembled body that cannot be manipulated.

The current input version uses actual RGB scenes, neutral instructions, and anonymous part/position handles. It supplies one acceptance target: a stage target for static tasks, or the complete required object/module for dynamic tasks.

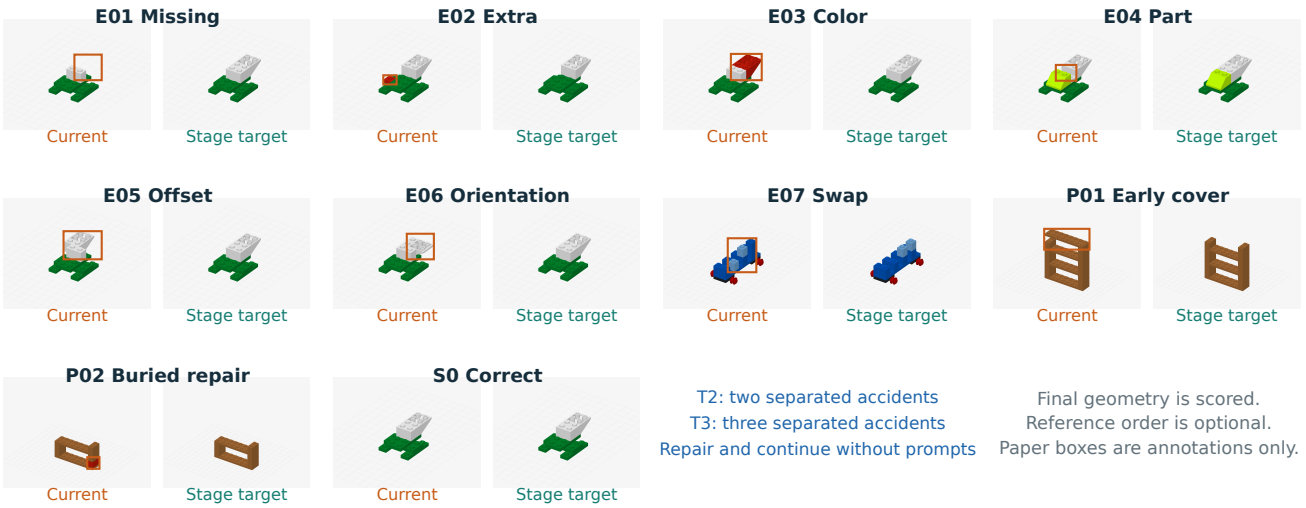

Figure 3. **Seven physical error labels, two process burdens, and a correct control.** Each pair shows an actual current state and its sole stage target with matched framing. Small helicopter examples use the required intermediate stage, not the completed nine-part object. E07 uses the longer bus; P01/P02 use the bookshelf. Orange boxes highlight pixel differences for the paper only. Simultaneous errors and temporally repeated events are different annotations.

Future manual pages are references, never additional acceptance goals. Isometric, rear, and top inspection are available. Current and target comparison panels each use an 800×600 scene on the same 1600×636 canvas. Full-object context remains visible for named modules. Manuals are queryable in community, expanded, and complete-LDraw tasks; legacy local static tasks have no step manual. Different information conditions are reported separately.

The response is one direct JSON action object, with no track-specific outer wrapper:

$$a_t \in \{\text{attach}(p, c), \text{detach}(p), \text{inspect}(q), \text{finish}\}. \quad (1)$$

Each manipulation moves one brick. Inspections query a supported view, part, candidate preview, or manual page. A preview is hypothetical and neither installs a part nor marks it as correct. The public observation excludes task family, hidden states, event schedule, and reference actions. Inventory is a part catalogue with spares, not a live remaining-piece oracle.

4.2. Legality and order-independent completion

An attachment must have a valid pose, a supported compatible connection, no endpoint collision, and a clear straight insertion path. Detachment checks a straight extraction path. Rejected attempts leave the scene unchanged and return a legality code. They consume the declared attempt budget, but do not prevent later success. Rigid pose/connection checks do not simulate force, slipping, flexible joints, or a gripper.

Let $\sigma(s)$ be the multiset of installed shape, color, and pose records, using only supported symmetry equivalences. Completion requires

$$C(\tau) = \mathbf{1}[a_T = \text{finish}]\mathbf{1}[\sigma(s_T) = \sigma(g)]. \quad (2)$$

Interchangeable same-shape, same-color pieces are not distinguished by internal ID. Asymmetric or decorated parts are not granted arbitrary rotational equivalence. The immutable body is excluded from part scores; adding ten roof pieces must not receive credit for 500 supplied wall pieces. No arbitrary global registration is applied. Floating pieces, duplicates, collisions, or state replacement cannot bypass execution.

A manual supplies one verified witness, not mandatory intermediate states. Two legal prefixes can have equal progress and different geometry (Fig. 5). Static repair ends at its stated stage even when later manual pages describe the finished object.

4.3. Interventions and visual qualification

Hidden edits occur between observations after accepted manipulations. The executor logs their actual timing, location, and legal primitives. Adaptive module selectors search the actual installed state; some older conditions use template-dependent opportunities. Neither a scheduled fault nor a nominal step threshold proves exposure. An infeasible event is deferred or untriggered. Multi-event tasks record every occurrence; one of two planned accidents is only partial exposure.

All 5,922 retained conditions pass fresh legal reference replay, with zero failures. This establishes witnessed feasibility, not visual understanding. The three complete LDraw

Figure 4. **Seven-part manual as one feasible route.** Each column shows one physical part and the resulting reference state. The seaplane body is supplied. Orange outlines explain the added part in the manual; ordinary current-state images do not highlight errors. These publication crops explain the module and do not replace the full-context evaluation views.

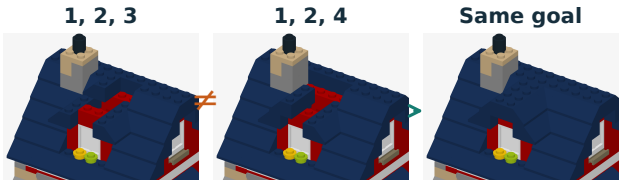

Figure 5. **Different legal prefixes, same goal.** These roof routes differ after three placements. Reference indices identify intended placements, not compulsory action times. Both routes are accepted by terminal scoring.

objects have 46 checked conditions: 36 reference fault images exceed 128 changed pixels at 24 intensity levels in at least one native view, and ten are controls. Adaptive locations require their own check. Older cropped module pixel checks do not certify the new full-context views. Complete current-interface image admission and independent human review remain pending. Low contrast and an event that cancels the latest action are flagged separately; absence of an obvious before/after change is not proof of a model miss.

5. Evaluation Metrics

5.1. Static correction and collateral damage

We distinguish S0 (260 correct-stage acceptance tasks) and S1 (3,437 repair tasks). Let I, F, G be the initial, final, and goal movable-part multisets and $H = I \cap G$ the initially correct placements. Exact SR is Eq. 2 averaged over valid S1 runs. Partial quality and repair progress are

$$\mathrm{F1}_{\mathrm{place}} = \frac{2|F \cap G|}{|F| + |G|}, \tag{3}$$

$$d(A, G) = |A - G| + |G - A|, \tag{4}$$

$$\mathrm{ER} = 1 - \frac{d(F, G)}{d(I, G)}, \qquad \mathrm{DR} = \frac{|H - F|}{|H|}. \tag{5}$$

ER can be negative. A wrong-color substitution contributes one missing correct placement and one excess placement; residual count is neither human error count nor minimum action cost. Temporarily dismantling then restoring a correct cover causes no terminal DR. On S0, false repair (FR) records any accepted attach/detach, even if the agent later restores the target. S0 accuracy and S1 SR are not pooled. Undefined denominators are N/A; F1 of two empty multisets is one. Attribute-combination F1, original missing recovery, and original excess clearance are supplemental.

5.2. Dynamic construction, exposure, and qualified recovery

D0 has 63 normal builds; D1 has 1,649 single-event tasks; D2 has 43 repeated-event tasks. BCR is normal-build completion and ICR is raw completion of valid intervention-assigned runs, including untriggered runs. Trigger rate TR counts actual events over planned events. Neither ICR nor TR is detection accuracy.

For causal comparisons, save a common environment/conversation prefix and branch it into control and intervention continuations. Both branches retain the same prior images, handle mapping, used budget, and settings. Let Q contain pairs with valid branches, a verified shared prefix, all planned events triggered, and verified incident effects. An edit that removes the agent's already-wrong piece cannot automatically earn repair credit. An early correct cover requires evidence that it blocks future access rather than an increase in terminal residual alone. For control/fault outcomes c_i, f_i, define

$$\mathrm{CRR} = \frac{\sum_{i \in Q} c_i f_i}{\sum_{i \in Q} c_i}, \qquad \Delta\mathrm{SR} = \frac{1}{|Q|} \sum_{i \in Q} (c_i - f_i). \tag{6}$$

CRR conditions on successful normal construction; the signed paired difference does not. Report planned/valid pairs, failed controls, untriggered/partial-event pairs, verified effects, and eligible N. Unreached faults are not repair failures or missed detections.

Recovery while continuing to build. For event states B before and A after, event-specific obligations are lost correct placements $L = (B \cap G) - A$ and incremental excess $X = (A - G) - (B - G)$. A later state repairs

Table 3. **E1: static repair (unmeasured).** Local-static panel; repeat the same schema separately for whole-object and named-module stages. SR/F1/ER/DR use S1; FR uses S0. Source- and object-macro estimates, valid counts, and coverage accompany results. Dashes are unmeasured, never zero.

Model	SR ↑	Place F1 ↑	ER ↑	DR ↓	FR ↓	Valid N
GPT-6.1 Sol	—	—	—	—	—	—
Claude Sonnet 5.5	—	—	—	—	—	—
Gemini 3.8 Flash	—	—	—	—	—	—
Claude Haiku 5.5	—	—	—	—	—	—
Qwen3.5-9B	—	—	—	—	—	—
Qwen3.5-27B	—	—	—	—	—	—
InternVL3.5-8B	—	—	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—	—	—
Gemma 3-12B	—	—	—	—	—	—

Table 4. **E2: dynamic construction and recovery (unmeasured).** W/M: whole-object/named-module scope. BCR uses D0; CRR requires normal success and verified complete exposure; Δ uses eligible pairs without filtering control success. Report D1 and D2 separately. ICR, TR and eligibility counts are retained in E3/E6/E9 and the ledger.

Model	W BCR	W CRR	W Δ	M BCR	M CRR	M Δ
GPT-6.1 Sol	—	—	—	—	—	—
Claude Sonnet 5.5	—	—	—	—	—	—
Gemini 3.8 Flash	—	—	—	—	—	—
Claude Haiku 5.5	—	—	—	—	—	—
Qwen3.5-9B	—	—	—	—	—	—
Qwen3.5-27B	—	—	—	—	—	—
InternVL3.5-8B	—	—	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—	—	—
Gemma 3-12B	—	—	—	—	—	—

these quantities without having to equal B; correct additional progress is allowed. The window ends before the next external event. Record first recovery in decisions, window-end recovery, and retention over at least two subsequent decisions. Observed regression fails retention; an otherwise correct short window is N/A. Process-only obstruction with empty L, X receives no automatic zero-latency recovery. Repair@k ($k = 2, 4, 8$) is reserved for sufficiently observed single-event or checkpoint trials.

Failures, aggregation, and missing data. Premature finish and model decision-budget exhaustion are ability failures. Final service errors, harness errors, and external stops are censored from ability rates and listed separately. A transient error that recovers within the same decision does not censor a completed run; its cost remains. Report valid coverage and completion per planned attempt alongside ability scores. Parsing, schema/ID errors, and physical rejections have separate counters. Legal-action rate uses well-formed manipulation attempts. Action cost is relative to a feasible witness, not SPL or proven optimal efficiency.

Normalize aliases, average within object, and report object- and source-macro estimates plus task-micro references. Separate input version, track, and task type. Error rows are multilabel; their counts do not add to a dataset size. Future 95% intervals resample source groups, not nominally independent variants. The current 154 groups have zero cross-split overlap; 45 displayed/debugged sources are public development. There is no untouched new-source community test in this release.

6. Diagnostics and Experiment Design

6.1. Reference replay and route sensitivity

The current replay audit covers 3,292 S, 1,712 D, 96 M, 776 expanded, and 46 original-object conditions, including aliases. These are legal reference executions, not model scores. A separate historical diagnostic uses twelve modules, two verified routes, and four conditions per route: 96 route–condition trials. Exact-prefix opportunities on alternative routes occur in 9/36 fault trials, versus 36/36 actual-state events (Table 5). Alternative routes were selected as

Table 5. **Historical route diagnostic.** All normal routes are legal and finish. Fault denominators are three families per module. No learned model is involved.

Check	Reference	Alternative
Normal terminal geometry	12/12	12/12
Every prefix matches manual	12/12	0/12
Exact-prefix opportunity	36/36	9/36
Actual-state event	36/36	36/36

regression counterexamples, not sampled from agent behavior. The blind continuation finishes 24/24 controls and 0/72 faults; a privileged inverse-based recovery finishes 96/96. Both are scripts. These historical numbers do not establish current-suite visual or model performance.

6.2. Planned model, human, and input comparisons

Tables 3–4 retain ten previously selected candidates: four API models and six open visual models, including Qwen2.5-VL and Gemma 3. The roster is a planned comparison; exact served revisions, image support, and account access must be checked before execution. Each model/condition runs once, with seed 41 for handle randomization. The candidate pool and audited test assignments are specified in the supplement. No old 467-question or 168-fault sample is silently reused as the current unified test split. The current existing-source test contains 392 main candidates: 309 local-static and 83 whole-object dynamic tasks. Public module and original-object results remain development analyses. Longer-community, repeated-event, and untouched-source coverage should be evaluated separately, with admission and eligibility counts for each claim.

All compared models receive the same declared task list, information, and budgets within a track. A proposed outer cap is $\max(40, 6N + 20)$ decisions, including inspection and malformed replies, while applicable native manipulation caps remain recorded. These proposed comparison settings are not a claim that every API adapter is already connected. Store every request, image hash/dimension, reply, accepted action, termination status, event, token use, and charge. Actual budgets and costs must come from a current-interface pilot, not old car prices.

The default condition is RGB with neutral task text and action handles, not pure image input. Text-state and RGB-plus-state conditions are planned separately: explicit poses can expose occluded geometry and are privileged relative to RGB. Free coordinates add localization burden and require a separate output experiment. Manual/history/view/feedback changes each hold the other factors fixed. Input changes require fresh paired prefixes and visibility checks.

Table 6. **E4: input controls (unmeasured).** Sol, Flash, and Qwen 27B are preselected diagnostic candidates. Report track-specific SR/BCR/CRR and counts. Manual removal applies only where a manual exists. Explicit state is privileged.

Condition	Sol	Flash	Qwen
Default RGB interface	—	—	—
No previous scene	—	—	—
No manual (supported tasks)	—	—	—
No candidate preview	—	—	—
Reduced legality feedback	—	—	—
Explicit state assistance	—	—	—

Table 7. **E3: paired outcomes (unmeasured).** Repeat for each model, dynamic scope, and D1/D2. n_{cf} counts control/fault success. CRR uses $n_{11}/(n_{11} + n_{10})$; signed Δ uses all four cells. Only prefix-, exposure-, and effect-qualified pairs enter.

Scope	n_{11}	n_{10}	n_{01}	n_{00}
Whole / D1	—	—	—	—
Module / D1	—	—	—	—
Whole / D2	—	—	—	—
Module / D2	—	—	—	—

The supplement reserves nineteen result tables covering correction, repeated events, checkpoint continuation, input/output, independent snapshot detection, human operation, efficiency, and service/format failures. Checkpoint tasks and detection/localization panels do not yet have independent task banks. The human plan uses graphical actions, not prose or JSON; historical agreement reports are not three independent submissions. No new model API calls or human performance measurements are reported in this revision.

7. Limitations and Conclusion

BRICKPLAN currently provides a program-checked candidate collection and an explicit measurement contract. Full-context image qualification, independent human operation, paired API execution, source-clustered intervals, and formal model results remain necessary. Only three whole objects are original complete LDraw files; many others are simple procedural objects. G3 contains just 52 main tasks, mainly longer procedural constructions. Community long modules and untouched source sets remain sparse. Public source files may also occur in model training.

Finite candidate poses, geometric connections, and straight manipulation paths simplify real assembly. Context is immutable; unsupported context geometry uses conservative obstacle bounds. Articulation, flexible parts, force stability, and robot control are outside scope. Pixel thresholds

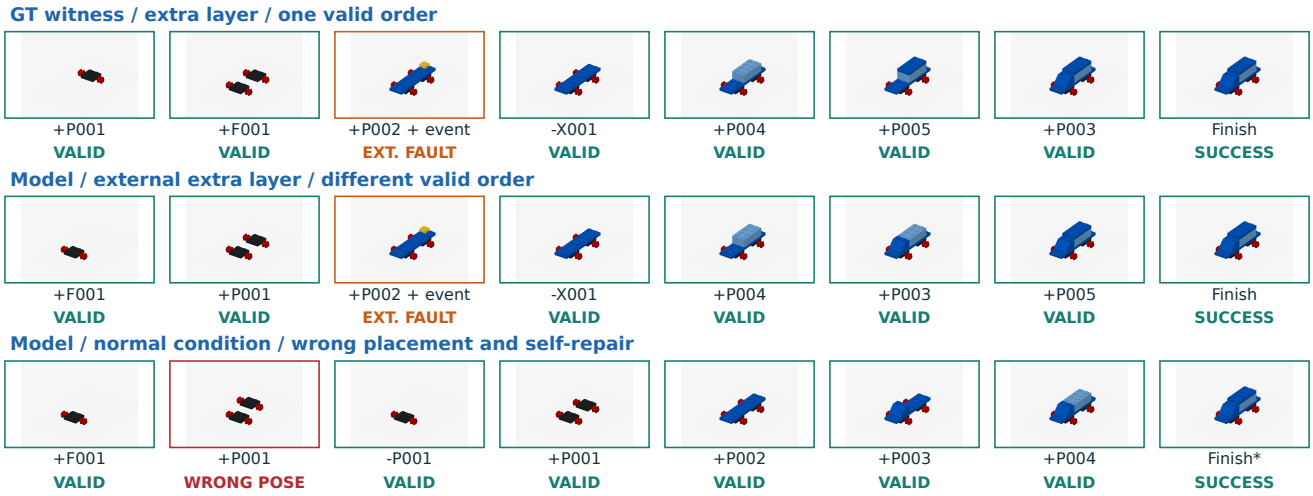

Figure 6. **Reference and archived model repair, eight states per row.** Top: one feasible GT witness. Middle: an archived Gemini 4 Argon rollout repairs an extra-layer event using another legal order. Bottom: a separate normal rollout corrects its own legal but wrong placement. All three eventually succeed. Green/orange/red indicate valid state/external change/wrong placement. +/− mean attach/detach; * omits intervening operations. The 32 model decisions replay exactly. These v5 examples are historical development illustrations, not current scores or shared-prefix pairs.

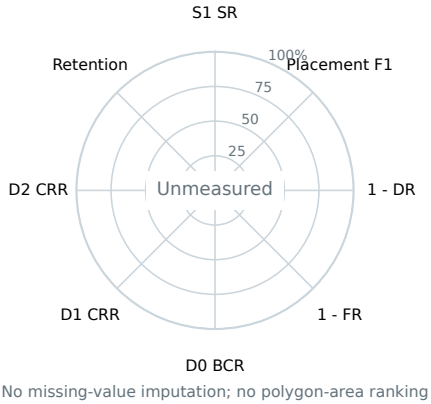

Figure 7. **Metric profile template, unmeasured.** Axes expose distinct outcomes; $1 - \mathrm{DR}$ and $1 - \mathrm{FR}$ reverse the two lower-is-better rates. No model curve, missing-value imputation, or polygon-area score is used. Track, counts, and uncertainty remain in tables.

cannot substitute for human readability. A failed episode cannot by itself establish a missed visual anomaly.

Within these limits, the benchmark separates static correction from ongoing assembly, accepts alternative legal routes, and makes fault exposure and recovery eligibility explicit. The current dataset, metrics, reproducible diagnostics, and reserved experiment tables form the basis for the remaining empirical study.

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